

Bioinformatics

Lecture 20

FASTA Algorithm

- The FASTA algorithm is a heuristic method for string comparison.
- It was developed by Lipman and Pearson in 1985 and further improved in 1988.
- It performs Local alignment of sequences.
- Definition: A heuristic method is a method that uses hash coding in which the sequence is broken into small “words or K-tuples” of specific sizes.
- Dynamic programming methods consume time to analyze and search the entire database.
- Heuristic approach overcomes this drawback.

Dynamic Programming – Problem

- Problem of Dynamic Programming

D.P. compute the score in **a lot of useless area** for optimal sequence

	G	A	A	T	T	C	A	G	T	T	A
G	1	1	1	1	1	1	1	1	1	1	1
G	1	1	1	1	1	1	1	2	2	2	2
A	1	2	2	2	2	2	2	2	2	2	2
T	1	2	2	3	3	3	3	3	3	3	3
C	1	2	2	3	3	4	4	4	4	4	4
G	1	2	2	3	3	4	4	5	5	5	5
A	1	2	3	3	3	4	5	5	5	5	6

- FASTA focuses on **diagonal area**

FASTA Algorithm – Steps

- Given a Query Sequence of either DNA or protein, the existing database of sequences from several genomes is searched.
- The Steps involved are as shown in FASTA algorithm flowchart (Fig 3) & as described below:
- It is based on a heuristic approach with restrictions of word size and window size.
- Create Hash-table of words from Query-sequence.

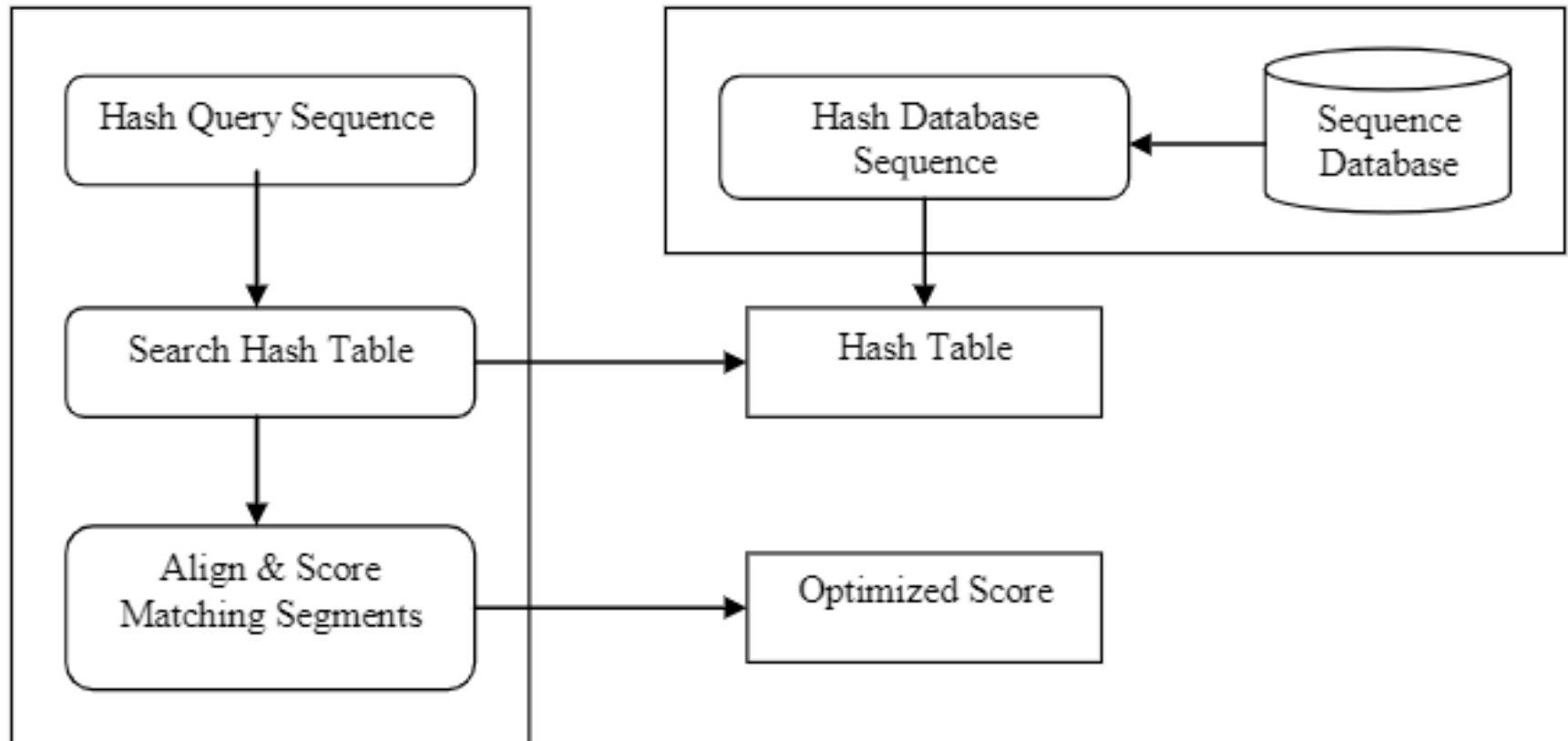
FASTA Algorithm – Steps

- A Hash-table is generated based on the Hashing-function, which maps the words to array position.
- As stated above, the word size could be 4 to 6 alphabets for nucleic acid sequences & 1 to 2 for amino acid sequences.
- The sequence characters are then compared to those in the database, which has been previously processed into the words of same length.
- FASTA uses the Blosom50 substitution matrix to score the top-10 alignments (without gaps) that contain the most similar words.
- These words are then merged into a gapped alignment, which is scored, producing an optimized score.

FASTA Algorithm – Steps

- FASTA produces an expectation score E – the expected number of random-alignments with z-scores greater than or equal to the value observed.
- This provides an estimate of statistical significance of the results.
- If E value is <0.02 , the similarity measure between the query sequence and a sequence(s) in a database is statistically significant.
- The z-score is derived from the optimal score after correcting for differences in sequence length.

FASTA Algorithm – Flowchart



FASTA Algorithm – Example

- Query Sequence: **WATSONCRICK**
- Target Sequence: **BALLCRICKET**
- **Step-1: Build query table (Indexing)**
- **Step-2: Build hash table for query string**
- **Step-3: Build target table (Indexing)**
- **Step-4: Build offset table**
- **Step-5: Build final table**

FASTA Algorithm – Example

STEP - 1:

W	A	T	S	O	N	C	R	I	C	K
0	1	2	3	4	5	6	7	8	9	10

FASTA Algorithm – Example

STEP - 1:

W	A	T	S	O	N	C	R	I	C	K
0	1	2	3	4	5	6	7	8	9	10

STEP - 2:

A	C	I	K	N	O	R	S	T	W
1	6, 9	8	10	5	4	7	3	2	0

FASTA Algorithm – Example

STEP - 1:

W	A	T	S	O	N	C	R	I	C	K
0	1	2	3	4	5	6	7	8	9	10

STEP - 2:

A	C	I	K	N	O	R	S	T	W
1	6, 9	8	10	5	4	7	3	2	0

STEP - 3:

0	1	2	3	4	5	6	7	8	9	10
B	A	L	L	C	R	I	C	K	E	T
-	0	-	-	2, 5	2	2	-1, 2	2	-	-8

FASTA Algorithm – Example

STEP - 3:

0	1	2	3	4	5	6	7	8	9	10
B	A	L	L	C	R	I	C	K	E	T
-	0	-	-	2, 5	2	2	-1, 2	2	-	-8

FASTA Algorithm – Example

STEP - 3:

0	1	2	3	4	5	6	7	8	9	10
B	A	L	L	C	R	I	C	K	E	T
-	0	-	-	2, 5	2	2	-1, 2	2	-	-8

STEP - 4:

-8	-1	2	5	0
1	1	5	1	1

FASTA Algorithm – Example

STEP - 4:

-8	-1	2	5	0
1	1	5	1	1

FASTA Algorithm – Example

STEP - 4:

-8	-1	2	5	0
1	1	5	1	1

- Maximum offset score is 5 of index: 2
- Add 2 Gaps to shift Target Sequence from LEFT side
- This will result in 5 matches.
- If index is Positive then add 2 gaps from left side
- If index is negative then add 2 gaps from right side

STEP - 5:

W	A	T	S	O	N	C	R	I	C	K	-	-
-	-	B	A	L	L	C	R	I	C	K	E	T

FASTA Algorithm – Example

STEP - 4:

-8	-1	2	5	0
1	1	5	1	1

- Maximum offset score is 5 of index: 2
- Add 2 Gaps to shift Target Sequence from LEFT side
- This will result in 5 matches.
- If index is Positive then add 2 gaps from left side
- If index is negative then add 2 gaps from right side

STEP - 5:

W	A	T	S	O	N	C	R	I	C	K	-	-
-	-	B	A	L	L	C	R	I	C	K	E	T